SNEHA SARAVANAN

2+ Years experience as Software Engineer | MS in Carnegie Mellon University | Raw data to production ready systems
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EDUCATION

Carnegie Mellon University, Pittsburgh, PA, USA December 2026
Master of Information Systems - Business Intelligence and Data *Analytics*

- Relevant Coursework: Machine Learning for Problem Solving, AI Engineering, Introduction to Deep Learning, Distributed Systems, Unstructured Data Analytics, Data Focused Python, Database Management, Object Oriented programming with Java

PES University, Bengaluru, KA, India May 2024
Bachelor of Technology in Computer Science

- Relevant Coursework: Data Structures & Algorithms in C++, Machine Intelligence, Operating Systems, Computer Architecture

SKILLS

Languages: Python, C++, SQL, Java, R
ML/ DL Frameworks: PyTorch, TensorFlow, Scikit-learn, XGBoost, Neural Networks, Drift Detection, A/B Testing,
LLM Frameworks: LangChain, RAG, PEFT, Groq, Llama, Sentence Embeddings
MLOPs/ Systems: AWS (Bedrock, SageMaker, S3, Redshift), Docker, Kubernetes, Airflow, Kafka, Spark, GitHub Actions, Prometheus, Hadoop, MLFLow, Grafana
Data & Visualization: Pandas, NumPy, Plotly, Matplotlib, Seaborn, time series forecasting

WORK EXPERIENCE

Software Engineer - Data & ML| Eltropy, Bengaluru, India July 2023 - August 2025

- Achieved 92% forecast accuracy by spearheading a hybrid SARIMA + Prophet staffing model on AWS SageMaker monitored via CloudWatch, replacing manual workforce allocation planning for 100+ credit union
- 89% validation accuracy on production loan forecasting by deploying a 5-layer LSTM (TensorFlow, Python) trained on 11 parameters to AWS Bedrock via a full Airflow-orchestrated ETL pipeline from S3 and Redshift
- Orchestrated 40% reduction in data quality failures by designing and implementing an automated validation framework on AWS Glue with 11 KPI checks and A/B testing across 100+ production tables before data reached downstream models
- Led 7× feature adoption growth (10 to 70+ customers) by analyzing underutilized product features via complex analytical SQL on Redshift, directly shaping roadmap prioritization decisions
- Zero downtime large-scale ETL pipelines by enforcing data contracts across the AWS ecosystem using Spark and Hadoop, eliminating schema-drift failures in distributed workflows

Research Assistant | Carnegie Mellon University, Pittsburgh, PA, USA October 2025 - Present

- Delivered full analyst self-service by consolidating 15+ datasets (200GB) into a unified Python/SQL framework, reducing data access latency from days to minutes
- Reduced data preparation time ~60% by building reproducible ML pipelines (feature engineering, preprocessing, validation) adopted across 10+ researchers, standardizing the team's full modeling workflow
- Cut pipeline failures ~40% by implementing automated schema and drift validation with Power BI dashboards, eliminating ad-hoc data fixes and enabling team-wide self-serve reporting

PROJECTS

RedisLite – In Memory Key-Value Store | C++ / Systems Programming

- Achieved high-throughput (100K+ ops/sec) low-latency (<1ms) performance by building an Redis-like Key-Value store in C++ with epoll driven concurrency and custom hash indexing
- Reduced 35% memory overhead via a custom arena allocator and optimized read-heavy workloads using shared mutex based concurrency control with LRU caching
- Ensured 0 data loss and predictable expiration via append-only persistence and min-heap TTL eviction, validated end-to-end under benchmarked load

Mytorch - Custom Deep Learning Library from Scratch & Speech Recognition | Deep Learning

- 89% phoneme classification accuracy with zero ML dependencies by building a full deep learning library from scratch in NumPy (PyTorch-inspired) and validated it on a speech recognition task end-to-end
- Full DL stack implemented from scratch – ReLU, Sigmoid, Tanh, GELU activations, MSE, Cross-Entropy losses, SGD, Adam optimizers, and batch normalization, with no external frameworks

Movie Recommendation System | ML in Production

- Eliminated 60% of model staleness by deploying an LLM recommendation service on Kubernetes with automated 72-hour retraining pipelines on rolling Kafka event streams
- 99.2% uptime maintained at scale by building an MLOps observability stack (Prometheus + Grafana) tracking accuracy drift, LLM inference cost, and pipeline health, with automated alerts cutting incident detection from hours to under 8 minutes
- 23% LLM licensing cost reduction by running A/B testing across 1 million users, confirmed statistically significant savings ($p < 0.05$, two-sample t-test) with <0.4 point rating degradation